



Sec 3 Express End-of-Year Paper 1 — Marking Scheme

Worked Solutions + Marking Scheme

Cambridge Learning Group · Sec 3 Express EOY · Paper 1 · 2026
15 pages · 90 marks

How to read this booklet:

BLACK — the question, written ON the original question paper

BLUE — the full step-by-step working and the reason for each line

RED — the marking scheme: where each B1 / M1 / A1 is earned, the traps,
and the key concept the question is testing

Mark codes: B1 independent answer mark · M1 method mark (earned even if

the answer is wrong) · A1 accuracy mark (only if the method mark was earned).

Answer all the questions.

- 1 The triangle DEF has an area of $\frac{1}{4}(7\sqrt{3}+9)$ cm². The length of DE is $(\sqrt{3}+2)$ cm and angle EDF is 60° . Find, without using a calculator, the length of DF , in cm, in the form $(a+b\sqrt{3})$ where a and b are integers. [3]

$$\text{Area} = \frac{1}{2}(DE)(DF) \sin 60^\circ$$

$$\frac{1}{4}(7\sqrt{3}+9) = \frac{1}{2}(\sqrt{3}+2)(DF) \times \frac{\sqrt{3}}{2}$$

$$7\sqrt{3}+9 = (3+2\sqrt{3})(DF)$$

$$DF = \frac{7\sqrt{3}+9}{3+2\sqrt{3}} \times \frac{3-2\sqrt{3}}{3-2\sqrt{3}}$$

$$DF = \frac{3\sqrt{3}-15}{-3}$$

$$\therefore DF = 5 - \sqrt{3} \text{ cm}$$

Key concept: area with the included angle, then rationalise.

M1 - $\frac{1}{2}(DE)(DF) \sin 60^\circ$ used, with

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

M1 - multiplying by the conjugate $3-2\sqrt{3}$

A1 - $DF = 5 - \sqrt{3}$, so $a=5$ and $b=-1$

Note - $(3)^2 - (2\sqrt{3})^2 = 9 - 12 = -3$ is negative.

Dropping that sign gives $\sqrt{3}-5$.

Trap - $\sqrt{3} \times \sqrt{3} = 3$, not $\sqrt{3}$.

- 2 Do not use a calculator in this question.

- (a) Simplify $\frac{7}{3-\sqrt{2}}$, giving your answer in the form $a+b\sqrt{2}$, where a and b are integers. [2]

$$\frac{7}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{7(3+\sqrt{2})}{9-2}$$

$$\frac{7(3+\sqrt{2})}{9-2} = \frac{7(3+\sqrt{2})}{7} = 3+\sqrt{2}$$

Key concept: multiply by the conjugate of the denominator.

M1 - conjugate $3+\sqrt{2}$ used and denominator

$$9-2=7$$

A1 - $3+\sqrt{2}$, so $a=3$ and $b=1$

- (b) Given that $(2+\sqrt{3})x = 5+4\sqrt{3}$, find the value of x in the form $a+b\sqrt{3}$, where a and b are integers. [2]

$$x = \frac{5+4\sqrt{3}}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$$

$$\text{denominator} = 4-3=1$$

$$x = 10 - 5\sqrt{3} + 8\sqrt{3} - 12$$

$$\therefore x = -2 + 3\sqrt{3}$$

M1 - conjugate $2-\sqrt{3}$ used

A1 - $x = -2 + 3\sqrt{3}$, so $a=-2$ and $b=3$

Note - the denominator rationalises to exactly 1 — the product must still be expanded.

- 3 It is given that $\sin A = \frac{-1}{\sqrt{5}}$ and $\tan B = \frac{3}{4}$, where A and B are in the same quadrant. Find the value of p if $\cos(A - B) = p\sqrt{5}$. [4]

$\sin A < 0 \Rightarrow A$ in quadrant 3 or 4
 $\tan B > 0 \Rightarrow B$ in quadrant 1 or 3
 same quadrant $\Rightarrow$ both in quadrant 3

$$\cos A = -\frac{2}{\sqrt{5}}$$

$$\sin B = -\frac{3}{5}, \cos B = -\frac{4}{5}$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\cos(A - B) = \left(-\frac{2}{\sqrt{5}}\right)\left(-\frac{4}{5}\right) + \left(-\frac{1}{\sqrt{5}}\right)\left(-\frac{3}{5}\right)$$

$$\cos(A - B) = \frac{11}{5\sqrt{5}} = \frac{11\sqrt{5}}{25}$$

$$\therefore p = \frac{11}{25}$$

Key concept: find the quadrant first; both ratios are then negative.

B1 - both A and B placed in quadrant 3

B1 - $\sin B = -\frac{3}{5}$ and $\cos B = -\frac{4}{5}$ from the right-angled triangle with sides 3, 4, 5

M1 - $\cos A \cos B + \sin A \sin B$ used

$$\mathbf{A1 - } p = \frac{11}{25}$$

Note - a CAST diagram alone earns the first B1 only — the ratios must be written down.

- (i) Express $y = -2x^2 - 12x + 5$ in the form $y = -a(x+b)^2 + c$, where a , b and c are constants. [2]

$$\begin{aligned} -2x^2 - 12x + 5 &= -2(x^2 + 6x) + 5 \\ -2(x^2 + 6x) + 5 &= -2[(x+3)^2 - 9] + 5 \\ -2[(x+3)^2 - 9] + 5 &= -2(x+3)^2 + 18 + 5 \\ \therefore y &= -2(x+3)^2 + 23 \end{aligned}$$

M1 - -2 factorised out and the square completed inside

A1 - $a=2, b=3, c=23$

- (ii) Hence find the range of values of the constant k for which the equation $-2x^2 - 12x + 5 = k$ does not have two distinct roots. [2]

$$-2(x+3)^2 + 23 = k$$

$$(x+3)^2 = \frac{23-k}{2}$$

two distinct roots need $\frac{23-k}{2} > 0$

none when $\frac{23-k}{2} \leq 0$

$$\therefore k \geq 23$$

Key concept: equal roots are not two distinct roots, so the boundary is included.

M1 - $b^2 - 4ac \leq 0$ or $(x+3)^2 \leq 0$ used

A1 - $k \geq 23$

Note - by discriminant: $2x^2 + 12x + (k-5) = 0$ gives $144 - 8k \leq 0$.

Trap - writing $k > 23$ loses the A1.

- 5 Show that the line $y = x - 1$ intersects the curve $y = nx^2 + 3x - n$ at two distinct points for all real values of n . [5]

$$x - 1 = nx^2 + 3x - n$$

$$nx^2 + 2x + (1 - n) = 0$$

$$\text{discriminant} = 2^2 - 4(n)(1 - n)$$

$$\text{discriminant} = 4 - 4n + 4n^2 = 4(n^2 - n + 1)$$

$$n^2 - n + 1 = \left(n - \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\left(n - \frac{1}{2}\right)^2 \geq 0, \text{ so } n^2 - n + 1 \geq \frac{3}{4} > 0$$

discriminant is positive for all real n , so the line meets the curve at two distinct points.

Key concept: show the discriminant is POSITIVE, not merely non-zero.

M1 - the two equations set equal

$$\text{A1 - } nx^2 + 2x + (1 - n) = 0$$

M1 - discriminant $4(n^2 - n + 1)$ formed

M1 - square completed, or discriminant of $n^2 - n + 1$ shown to be $-3 < 0$

A1 - concludes discriminant > 0 , hence two distinct points

Note - asserting positivity without justification scores the M marks only.

6 Solve the equation $\log_2(x-2) + 2\log_4(x-3) = \frac{1}{2}\log_3 9$. [5]

$$\log_3 9 = 2, \text{ so } \text{RHS} = \frac{1}{2} \times 2 = 1$$

$$2\log_4(x-3) = \frac{2\log_2(x-3)}{\log_2 4} = \log_2(x-3)$$

$$\log_2[(x-2)(x-3)] = 1$$

$$(x-2)(x-3) = 2$$

$$x^2 - 5x + 4 = 0$$

$$(x-1)(x-4) = 0$$

$$x = 1 \text{ or } x = 4$$

$$\text{need } x-2 > 0 \text{ and } x-3 > 0, \text{ so } x > 3$$

$$\therefore x = 4$$

Key concept: change base to 2, combine, then test the domain.

B1 - RHS = 1

M1 - $\log_4$ changed to base 2

M1 - logs combined and the equation made quadratic

A1 - $x = 1$ and $x = 4$

A1 - $x = 1$ rejected because $x > 3$; $x = 4$

Trap - rejecting with $x > 0$ is not a reason — $x = 1$ satisfies it.

- (i) Explain why $\frac{3x^2+3x+10}{(x+1)(x^2+4)}$ cannot be expressed in the form [2]
 $\frac{A}{x+1} + \frac{B}{x^2+4}$, where A and B are constants.

x^2+4 is an irreducible quadratic — $x^2 = -4$ has no real root, so it has no real linear factor. A factor of degree 2 needs a numerator of degree 1, i.e. $Bx+C$.

With only A and B there are 2 unknowns but 3 coefficients to match.

B1 - states x^2+4 is irreducible / has no real linear factors

B1 - states the numerator over x^2+4 must be linear, $Bx+C$

Note - writing the correct form without a reason earns the second B1 only.

- (ii) Express $\frac{3x^2+3x+10}{(x+1)(x^2+4)}$ in partial fractions. [4]

$$\frac{3x^2+3x+10}{(x+1)(x^2+4)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+4}$$

$$3x^2+3x+10 = A(x^2+4) + (Bx+C)(x+1)$$

$$x = -1: 10 = 5A, \text{ so } A = 2$$

$$\text{compare } x^2: 3 = A+B, \text{ so } B = 1$$

$$\text{compare constants: } 10 = 4A+C, \text{ so } C = 2$$

$$\therefore \frac{2}{x+1} + \frac{x+2}{x^2+4}$$

M1 - correct form with $Bx+C$ written down

M1 - $x = -1$ substituted to give $A = 2$

A1 - $B = 1$

A1 - $C = 2$

Note - check:

$$2(x^2+4) + (x+2)(x+1) = 3x^2+3x+10.$$

8 The coordinates of the points A , B and C are $(2,3)$, $(-1,-1)$ and $(3,1)$ respectively.

(i) Show that AC is perpendicular to BC . [3]

$$m_{AC} = \frac{1-3}{3-2} = -2$$

$$\text{M1} - m_{AC} = -2$$

$$m_{BC} = \frac{1-(-1)}{3-(-1)} = \frac{2}{4} = \frac{1}{2}$$

$$\text{M1} - m_{BC} = \frac{1}{2}$$

$$m_{AC} \times m_{BC} = -2 \times \frac{1}{2} = -1$$

A1 - product = -1 and the conclusion stated

$\therefore AC$ is perpendicular to BC .

(ii) Given that $ABCD$ is a parallelogram, find the coordinates of the point D . [2]

diagonals bisect:

midpoint AC = midpoint BD

M1 - midpoint AC = midpoint BD used

$$\text{midpoint } AC = \left(\frac{5}{2}, 2\right)$$

A1 - $D(6,5)$

$$\frac{-1+x_D}{2} = \frac{5}{2}, \frac{-1+y_D}{2} = 2$$

Trap - midpoint AB = midpoint DC is false — those are opposite sides.

$\therefore D(6,5)$ Find the area of the parallelogram $ABCD$. [2]

right angle at C , so area = $2 \times \frac{1}{2}(AC)(BC)$

M1 - a correct area method (perpendicular sides, or the shoelace formula)

A1 - 10 units²

$$AC = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

$$BC = \sqrt{4^2 + 2^2} = 2\sqrt{5}$$

$$\text{area} = \sqrt{5} \times 2\sqrt{5} = 10 \text{ units}^2$$

Note - part (i) makes $AC \perp BC$ available — that is the short route.

- 9 The curve $y = a \cos 2x + b$, where $a > 0$, has a maximum value of 4 and a minimum value of -10 , for $0 \leq x \leq 2\pi$.

(i) Find the value of a and of b . [3]

maximum: $a + b = 4$

minimum: $-a + b = -10$

adding: $2b = -6$, so $b = -3$

$\therefore a = 7, b = -3$

M1 - both $a + b = 4$ and $-a + b = -10$ formed

A1 - $b = -3$

A1 - $a = 7$

(ii) State the period and the amplitude of the curve $y = a \cos 2x + b$. [2]

$$\text{period} = \frac{2\pi}{2} = \pi$$

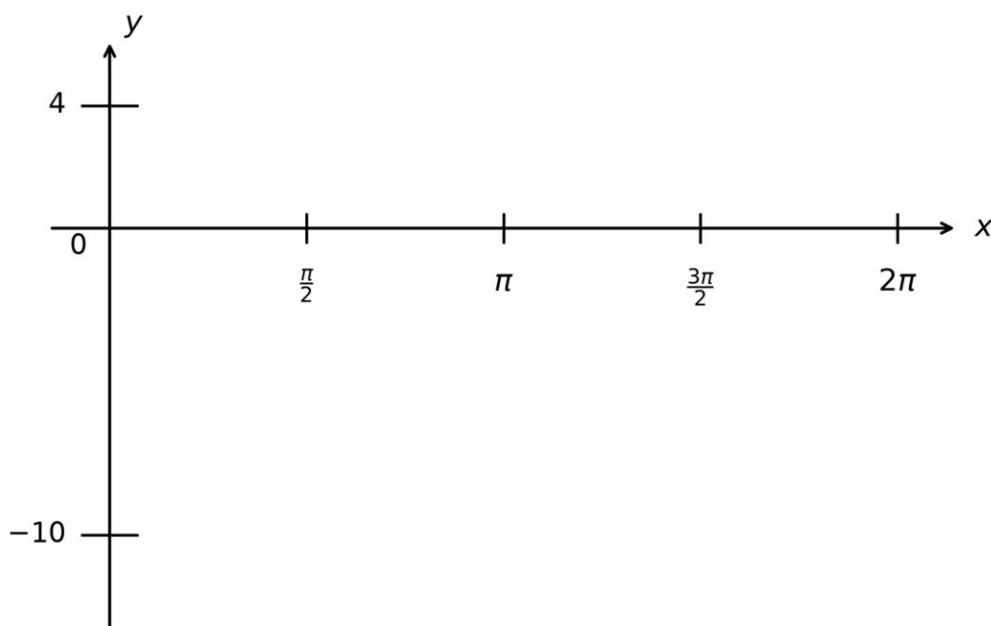
$$\text{amplitude} = 7$$

B1 - period π

B1 - amplitude 7

Trap - the amplitude is $a = 7$, not 14, and b does not change it.

(iii) Sketch the curve of $y = a \cos 2x + b$ for $0 \leq x \leq 2\pi$ on the axes below. [2]



$y = 7 \cos 2x - 3$: cosine shape, 2 complete cycles on $0 \leq x \leq 2\pi$

maximum 4 at $x = 0, \pi, 2\pi$; minimum -10

at $x = \frac{\pi}{2}, \frac{3\pi}{2}$

oscillates about $y = -3$, crossing it at

$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

B1 - correct shape, starting at a maximum, 2 equal cycles

B1 - maximum 4 and minimum -10 shown at the right values of x

- 10 Mr Goh bought an antique vase at the beginning of 2015 as an investment. According to the seller, the vase's value, \$ P , increases over time, t years, after purchase. The value of the vase is given by $P = 45\,000 e^{kt}$, where k is a constant. An evaluation at the beginning of 2016 valued the vase at \$50 000.

(a) Show that $k = 0.10536$, correct to 5 significant figures. [3]

$$t = 1, P = 50\,000:$$

$$50\,000 = 45\,000 e^k$$

$$e^k = \frac{50\,000}{45\,000} = \frac{10}{9}$$

$$k = \ln \frac{10}{9} = 0.1053605\dots$$

$$\therefore k = 0.10536 \text{ (5 s.f.)}$$

M1 - $t = 1$ and $P = 50\,000$ substituted

$$\text{M1} - e^k = \frac{10}{9}$$

A1 - $k = 0.10536$ correct to 5 s.f.

(b) Assuming that the value of the vase continues to increase at the same rate, calculate the year in which Mr Goh's investment becomes tripled. [4]

$$\text{tripled: } P = 3 \times 45\,000 = 135\,000$$

$$135\,000 = 45\,000 e^{kt}$$

$$e^{kt} = 3$$

$$t = \frac{\ln 3}{0.1053605} = \frac{1.098612}{0.1053605}$$

$$t = 10.427\dots \approx 10.43 \text{ years}$$

10.43 years after the start of 2015 falls in 2025.

Key concept: triple the ORIGINAL investment, then read off the calendar year.

$$\text{M1} - P = 3 \times 45\,000 = 135\,000$$

$$\text{M1} - e^{kt} = 3$$

$$\text{M1} - t = \frac{\ln 3}{k}$$

A1 - the year 2025

Trap - tripling the \$50 000 valuation, or stopping at $t = 10.4$ without naming the year.

(a) Find the term independent of x in the expansion of $3x^3\left(2x - \frac{1}{4x^2}\right)^{12}$. [3]

general term of $\left(2x - \frac{1}{4x^2}\right)^{12}$:

$$\binom{12}{r} (2x)^{12-r} \left(-\frac{1}{4x^2}\right)^r$$

i.e. $\binom{12}{r} 2^{12-r} (-1)^r 4^{-r} x^{12-3r}$

with $3x^3$ the power of x is

$$3 + 12 - 3r = 15 - 3r$$

$$15 - 3r = 0 \Rightarrow r = 5$$

$$\text{term} = 3 \binom{12}{5} 2^7 (-1)^5 4^{-5}$$

(b) In the expansion of $(2+mx)^{15}$, where m is a constant, the ratio of the coefficient of x^7 to the coefficient of x^9 is $\frac{4}{7}$. Find the possible values of m . [5]

coefficient of x^7 : $\binom{15}{7} 2^8 m^7$

coefficient of x^9 : $\binom{15}{9} 2^6 m^9$

$$\frac{\binom{15}{7} 2^8 m^7}{\binom{15}{9} 2^6 m^9} = \frac{4}{7}$$

$$\frac{6435}{5005} = \frac{9}{7}, \text{ so } \frac{9}{7} \times \frac{4}{m^2} = \frac{4}{7}$$

$$\frac{36}{7m^2} = \frac{4}{7}$$

$$m^2 = 9$$

$$\therefore m = 3 \text{ or } m = -3$$

Key concept: write the GENERAL term, then set the power of x to zero.

M1 - general term $\binom{12}{r} (2x)^{12-r} \left(-\frac{1}{4x^2}\right)^r$

M1 - $15 - 3r = 0$ giving $r = 5$

A1 - -297

Trap - $(-1)^5$ makes it negative; $\binom{12}{5} = 792$ and

$$\frac{128}{1024} = \frac{1}{8}.$$

M1 - coefficient of x^7 as $\binom{15}{7} 2^8 m^7$

M1 - coefficient of x^9 as $\binom{15}{9} 2^6 m^9$

M1 - ratio formed and equated to $\frac{4}{7}$

M1 - simplified to $m^2 = 9$

A1 - $m = \pm 3$

Trap - both signs are needed — the ratio depends on m^2 .

- (a) It is given that $f(x) = x^3 + x^2 + ax + 2b$ and $g(x) = x^3 - 4x^2 - ax - b$, where a and b are constants.

$(x+3)$ is a factor of $f(x)$.

$g(x)$ leaves a remainder of 42 when divided by $(x-5)$.

Find the values of a and of b .

[5]

$$(x+3) \text{ a factor} \Rightarrow f(-3)=0$$

$$-27+9-3a+2b=0$$

$$2b=3a+18 \dots(1)$$

$$g(5)=42: 125-100-5a-b=42$$

$$b=-5a-17 \dots(2)$$

$$(2) \text{ into } (1): 2(-5a-17)=3a+18$$

$$-13a=52$$

$$\therefore a=-4, b=3$$

$$\text{M1 - } f(-3)=0 \text{ used}$$

$$\text{A1 - } 2b=3a+18$$

$$\text{M1 - } g(5)=42 \text{ used}$$

$$\text{A1 - } b=-5a-17$$

$$\text{A1 - } a=-4 \text{ and } b=3$$

Trap - $(-3)^3 = -27$ and $(-3)^2 = 9$; a fractional a is the signal of a slip.

- (b) Solve the equation $x^3 - x^2 - 7x + 7 = 0$, expressing the roots in exact form. [5]

$$x^3 - x^2 - 7x + 7 = 0$$

$$x^2(x-1) - 7(x-1) = 0$$

$$(x-1)(x^2-7) = 0$$

$$x^2 = 7$$

$$\therefore x=1, x=\sqrt{7}, x=-\sqrt{7}$$

M1 - grouping, or the factor theorem with $x=1$

$$\text{A1 - } (x-1)(x^2-7)$$

$$\text{M1 - } x^2=7 \text{ solved}$$

$$\text{A1 - } x=1$$

$$\text{A1 - } x=\pm\sqrt{7}$$

Trap - exact form means $\pm\sqrt{7}$, not ± 2.65 .

- 13 A marine monitoring buoy transmits data via Wi-Fi to a receiving station on shore. As the signal travels over the sea, it experiences a loss in strength. The signal loss, L (in dB), depends on the distance, d m, from the buoy and can be modelled by the formula $L = A d^k$, where A and k are constants.

The table below shows corresponding values of d and L .

d	9	50	200	400	900
L	8	15	25	32	43

- (a) Explain how a straight line graph can be drawn to represent the formula, and draw it for the given data on the grid below. [4]

$$L = A d^k$$

$$\lg L = \lg A + k \lg d$$

$$\lg L = k \lg d + \lg A$$

Comparing with $Y = mX + c$: plot $\lg L$

(vertical) against $\lg d$ (horizontal).

The graph is a straight line of gradient k and vertical intercept $\lg A$.

$\lg d$: 0.954, 1.699, 2.301, 2.602, 2.954

$\lg L$: 0.903, 1.176, 1.398, 1.505, 1.633

Key concept: take logs of both sides to linearise a power law.

B1 - $\lg$ taken of both sides: $\lg L = \lg A + k \lg d$

B1 - compares with $Y = mX + c$; gradient k , intercept $\lg A$

M1 - both $\lg$ rows computed

A1 - points plotted correctly and a straight line drawn

(b) Use your graph to estimate the value of A and of k .

[4]

gradient $k = \frac{1.633 - 0.903}{2.954 - 0.954} = \frac{0.730}{2.000}$

$$k \approx 0.365$$

intercept: $\lg A = \lg L - k \lg d$

$$\lg A = 0.903 - 0.365(0.954) = 0.555$$

$$A = 10^{0.555}$$

$$\therefore A \approx 3.59, k \approx 0.365$$

Key concept: read two points off the **DRAWN LINE**, as far apart as it allows.

M1 - gradient from two points on the line

A1 - $k \approx 0.365$

M1 - intercept used to find $\lg A$

A1 - $A \approx 3.59$

Note - accept k from 0.35 to 0.38 and A from 3.3 to 3.9.

Note - check: $3.59 \times 200^{0.365} \approx 24.8$ against the tabulated $L = 25$.

(a) Prove the identity

[5]

$$\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{\cos \theta} = 2 \tan \theta$$

$$\text{LHS} = \frac{\cos^2 \theta - (1 - \sin \theta)^2}{(1 - \sin \theta) \cos \theta}$$

$$(1 - \sin \theta)^2 = 1 - 2 \sin \theta + \sin^2 \theta$$

$$\text{numerator} = \cos^2 \theta - 1 + 2 \sin \theta - \sin^2 \theta$$

$$\text{using } \cos^2 \theta = 1 - \sin^2 \theta:$$

$$\text{numerator} = 2 \sin \theta - 2 \sin^2 \theta = 2 \sin \theta (1 - \sin \theta)$$

$$\text{LHS} = \frac{2 \sin \theta (1 - \sin \theta)}{(1 - \sin \theta) \cos \theta} = \frac{2 \sin \theta}{\cos \theta}$$

$$\text{LHS} = 2 \tan \theta = \text{RHS}$$

Key concept: one common denominator, then
 $\cos^2 \theta = 1 - \sin^2 \theta$.

M1 - single fraction $\frac{\cos^2 \theta - (1 - \sin \theta)^2}{(1 - \sin \theta) \cos \theta}$

M1 - $(1 - \sin \theta)^2$ expanded

M1 - $\cos^2 \theta$ replaced by $1 - \sin^2 \theta$

M1 - numerator factorised as $2 \sin \theta (1 - \sin \theta)$

A1 - cancels to $2 \tan \theta$

Trap - combining as $\frac{\cos \theta - (1 - \sin \theta)}{\cos \theta (1 - \sin \theta)}$ **— the numerators must be cross-multiplied.**

(b) Hence, solve the equation

[4]

$$\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{\cos \theta} = \cot \theta \text{ for } 0 \leq \theta \leq 2\pi$$

$$\text{by (a): } 2 \tan \theta = \cot \theta = \frac{1}{\tan \theta}$$

$$2 \tan^2 \theta = 1$$

$$\tan \theta = \pm \frac{1}{\sqrt{2}}$$

$$\text{basic angle } \alpha = 0.6155 \text{ rad}$$

$$\tan \theta > 0: \theta = 0.6155, \pi + 0.6155 = 3.757$$

$$\tan \theta < 0:$$

$$\theta = \pi - 0.6155 = 2.526, 2\pi - 0.6155 = 5.668$$

$$\therefore \theta = 0.615, 2.53, 3.76, 5.67 \text{ rad}$$

M1 - result of (a) used: $2 \tan \theta = \cot \theta$

M1 - $\tan^2 \theta = \frac{1}{2}$

A1 - two correct values

A1 - all four values, in radians to 3 s.f.

Trap - the range $0 \leq \theta \leq 2\pi$ is in radians — degrees score at most 3.

- (c) Use the result in part (a) to find the set of values of the constant k for which there are no solutions to the equation [3]

$$\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{\cos \theta} = k(1 + \tan^2 \theta)$$

by (a) the equation becomes

$$2 \tan \theta = k(1 + \tan^2 \theta)$$

let $t = \tan \theta$:

$$k t^2 - 2t + k = 0$$

no real t needs discriminant < 0

$$(-2)^2 - 4(k)(k) < 0$$

$$4 - 4k^2 < 0$$

$$k^2 > 1$$

$$\therefore k < -1 \text{ or } k > 1$$

Key concept: the identity turns it into a quadratic in $\tan \theta$.

$$\text{M1 - } k t^2 - 2t + k = 0 \text{ with } t = \tan \theta$$

M1 - discriminant < 0 used

$$\text{A1 - } k < -1 \text{ or } k > 1, \text{ i.e. } k^2 > 1$$

Note - $k = 0$ gives $t = 0$, which does have solutions — consistent, since $0^2 < 1$.

----- End of Paper -----